Bentley iTwin Capture Modeler

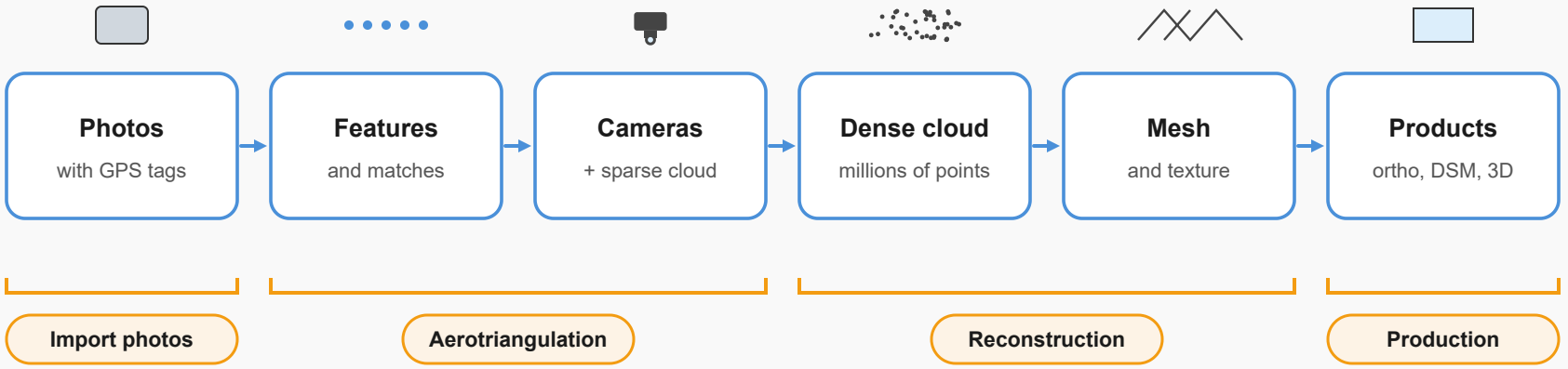
Processing Lab — step by step

Today

- Photos in, orthophoto and 3D model out
- Steps: project, import, aerotriangulation, reconstruction, production
- Then open the orthophoto in QGIS
- What to check along the way
- What to turn in

Six stages, four buttons

What structure from motion does, and what the software calls each part of it.



Bentley iTwin Capture Modeler, the four steps you click

The software does the arithmetic. The photos decide whether the arithmetic has anything to work with.

Six stages, four buttons. Import, Aerotriangulation, Reconstruction and Production map onto the pipeline you saw Tuesday.

Step 1 — Create a Project

- Open **iTwin Capture Modeler**
- Create a **new project**; name it and choose the project folder
- **New Capture**, then **New Block**: the set of photos
- Engine must be running, or jobs stay **pending**

Step 2 — Import Photos

- Open the block's **Photos** tab
- Add all the photos of a directory
- GPS positions are read from the photo metadata
- Check the block **3D view**: do the cameras match your flight?

The Bad Photo Hunt

Before the next step, scan your photos. **Delete** any that are:

- Blurry
- Pointed at the ground during takeoff
- Far from the study area

Garbage in, garbage out.

Step 3 — Aerotriangulation

- Click **Submit Aerotriangulation**
- Keep the **default settings** (Compute, Normal, Default)
- Click **Submit**, then wait
- Result: a new block with a **sparse point cloud**

Reading the AT result

- **Photos calibrated:** all, or nearly all
- **View quality report:** reprojection error under 1 px is good
- Sparse cloud, **3D view**: most points on brick and grass
- **Holes:** shiny surfaces, windows, fresh asphalt

Step 5 — Reconstruction

- (Step 4, ground control, is skipped in this lab)
- **New Reconstruction**: **Orthomosaic/DSM** for the map, **3D reconstruction** for mesh and points
- **Spatial framework:** set `EPSG:32612`, check the region of interest
- **Geometric precision:** Medium for the map

Step 6 — Production

Click **Submit new production**, pick the purpose:

- **Export Orthomosaic/DSM** — **TIFF/GeoTIFF**, `EPSG:32612`
- **Export 3D mesh** — **iTwin Ready** or **OBJ wavefront**
- **Export 3D point cloud** — **LAS**

QGIS needs the orthophoto GeoTIFF.

While it runs

- **Aerotriangulation is the brain.** If it fails, the model fails
- **Sparse** cloud comes first: points only
- **Dense** reconstruction, the actual model, comes after
- Look at the sparse cloud now: where are the holes?

The sparse cloud has a hole over the parking lot.

What happened on the flight?

What to turn in

The lab page has the deliverables: Data Processing.

Full written steps: Bentley iTwin.

Screenshots

Screenshots of each step will be added from the class project.